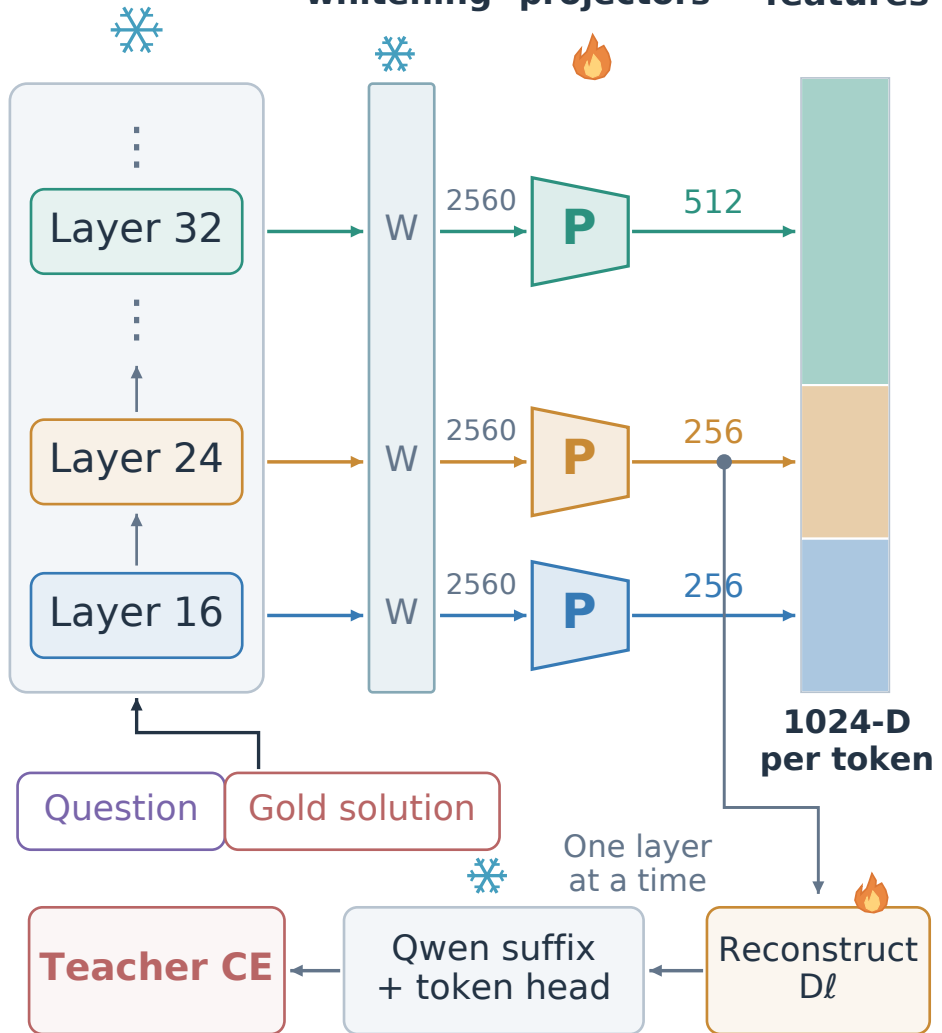


1 Representation learning

Qwen3-4B **Covariance whitening** **Learned projectors** **Concatenate features**



Freeze this representation for flow training.

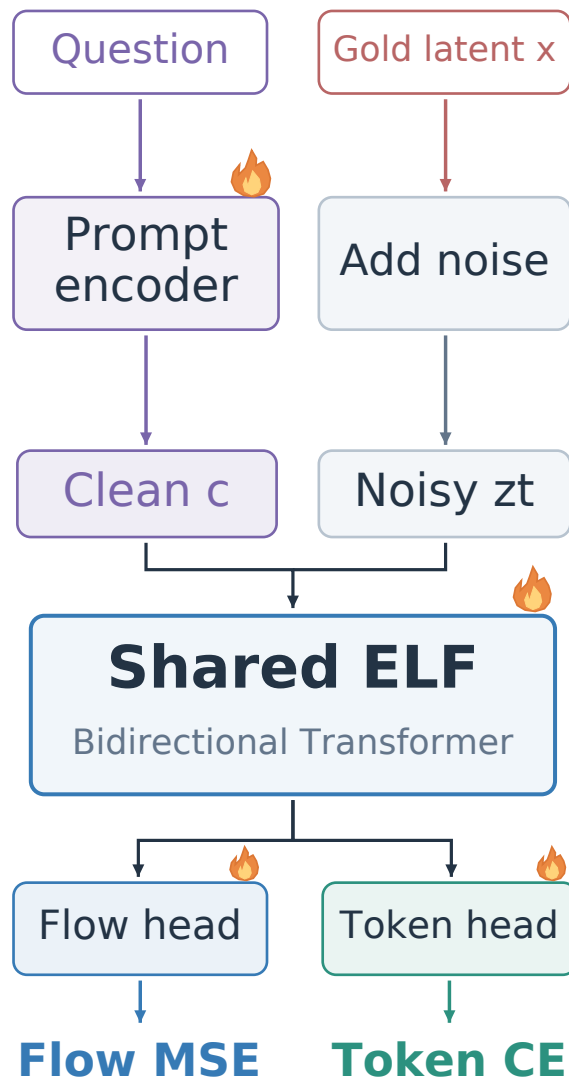


Frozen



Trained

2 Flow training

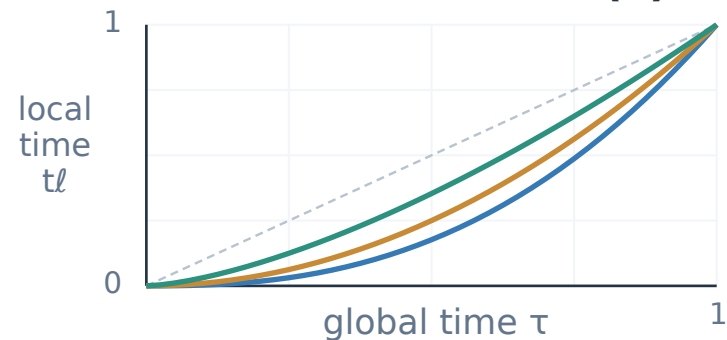


Teacher prompts $\rightarrow$ MSE
warm start $\rightarrow$ joint training

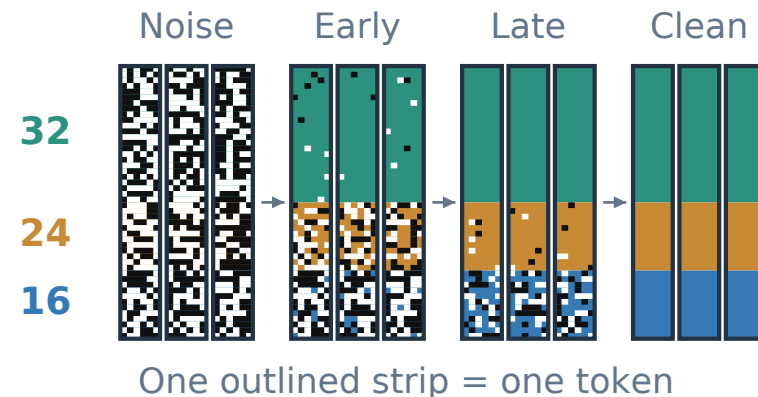
3 Async inference

Encode question once $\rightarrow$ cache c

Local clocks: $t_l = f_l(\tau)$



— $\gamma_{16}=2.5$ — $\gamma_{24}=2$ — $\gamma_{32}=1.5$



Joint ELF calls + Euler updates

ELF + token head $\rightarrow$ answer

Noise contrast exaggerated for illustration

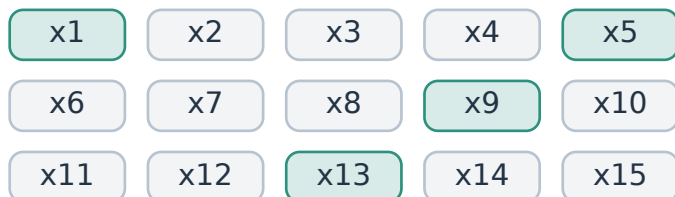
1 Collect endpoints

Question q

Old rollout policy
ELF + prompt encoder

32 steps · synchronous (1, 1, 1)

15 sampled latent endpoints



Decode text for correctness scoring

+ 1 reference solution
Cached gold Qwen latent

Retain generated terminal latents
for the NFT update.

2 Reward + selection

Correct generation **0.75**

Incorrect generation **0**

Gold reference **1**

Discard groups with
all 15 generations correct

Center rewards within each group;
scale by retained raw-reward SD

Advantage $A \rightarrow$ weight ρ
 $\rho = \text{clip}(\frac{1}{2} + A/10, 0, 1)$

Weights are detached.

Refresh the rollout policy from the current policy

3 NFT update

Latent endpoint x + fresh (t, ϵ)

Current policy θ, ϕ

Old policy · stop-grad

Fixed reference · stop-grad

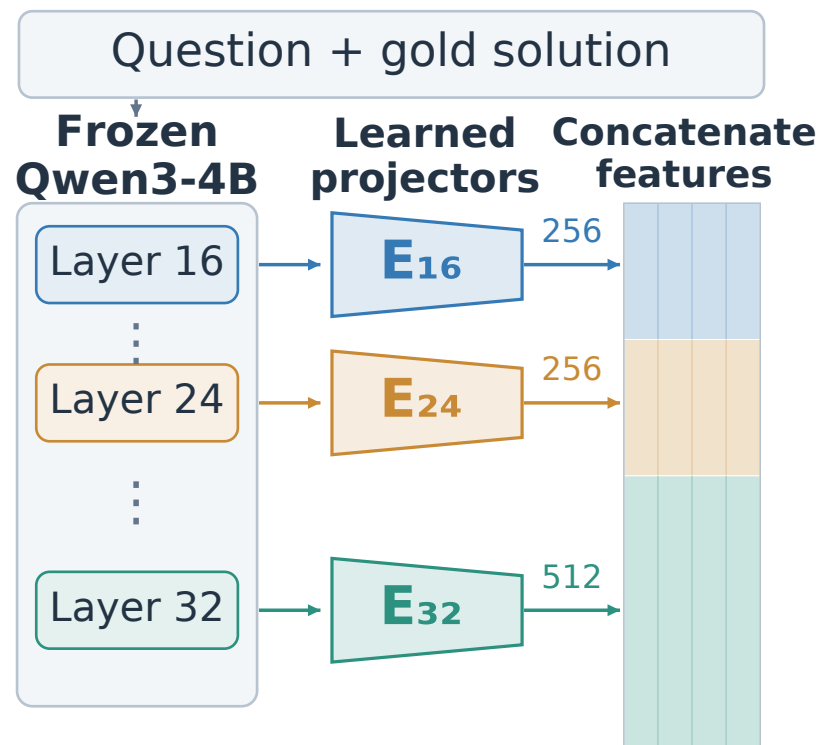
SCCFG velocity for each role
Detached guidance correction

Reward-weighted NFT loss
+ fixed-reference penalty

$$e+ = e_c \quad e- = 2e_o - e_c \quad (\beta = 1)$$

Update ELF + prompt encoder
Decoder-only weights frozen

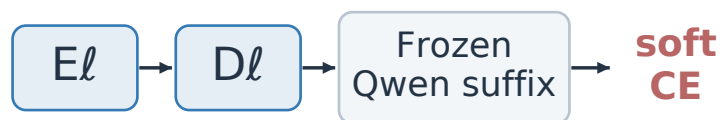
1 Representation learning



$$1024-D = 256 + 256 + 512$$

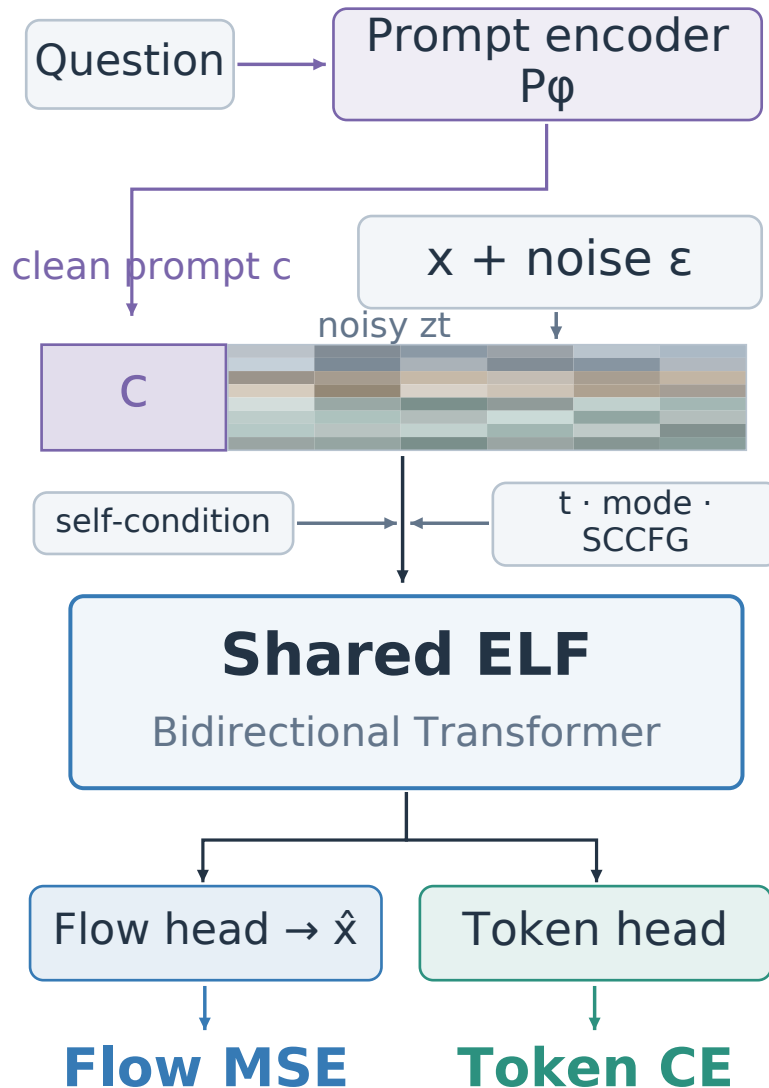
$$x_l = (h_l - \mu_l) E_l$$

Teacher-CE supervision



Covariance whitening; then freeze the representation for flow training.

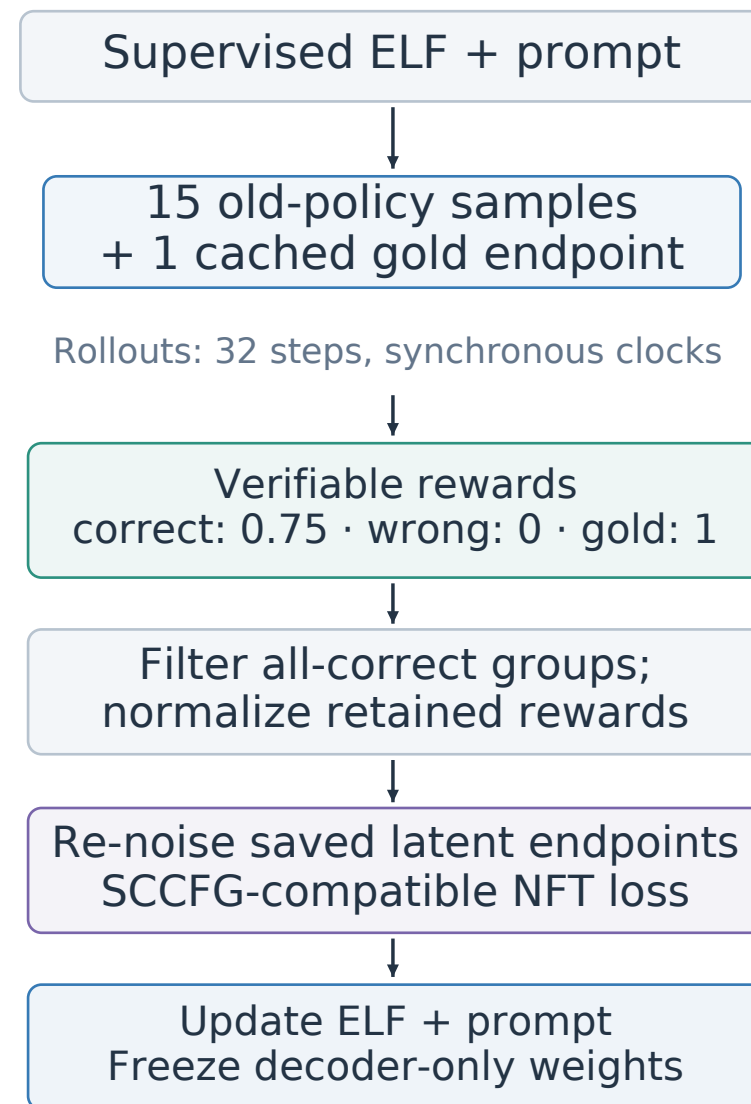
2 Flow training



80% flow rows · 20% decoder rows

Teacher prompt → MSE warm start → joint

3 NFT post-training



Forward-process training;
fixed reference-policy penalty